



Sustainable Development Verified Impact Standard

IMPROVED COOKSTOVE GROUPED PROJECT IN PAPUA NEW GUINEA



Tasman Environmental Markets

Project Title	Improved Cookstove Grouped Project in Papua New Guinea
Version	1.0
Date of Issue	XX-January-2023
Project Location	Papua New Guinea
Project Proponent(s)	Tasman Environmental Markets Dr. Gabriel Machovsky-Capuska, Head of Science and Sustainability TEM 15th Floor 31 Market Street, Sydney, NSW 2000, Australia Email: gabriel@tem.com.au

Project Lifetime	01 December 2023 to 31 November 2032; 10-year lifetime
History of SD VISTa Status	No previous attempts at SD VISTa certification made to date.
Other Certification Programs	VERRA Verified Carbon Standard
Expected Future Assessment Schedule	Where known, the expected schedule for initial verification, or, if using project design evaluation, validation or project implementation evaluation will be provided

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1 SUMMARY OF SDG CONTRIBUTIONS

Table 1: Summary of Project SDG Contributions

Row number	Estimated Project Contribution by the End of Project Lifetime	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Section Reference	Claim, Asset or Label
1)	The project activity will replace 100,000 traditional cookstoves with fuel efficient improved cookstoves (ICS). The efficiency of changing from a traditional cookstove to a more efficient one will result in the consumption of lesser firewood that is usually purchased at high prices. ¹ Hence reducing the economic burden of families under the project activity.	1.1	1.1.1 Proportion of population below the international poverty line, by sex, age, employment status, and geographical location (urban/rural).	Implement activities to decrease	3.2 #4	Claim
2)	The project will improve food security and nutrition status, particularly for children and women by reducing inadequate cooking, the burden of firewood collection, the time to prepare food, the cost to buy firewood.	2.1	2.1.1 Prevalence of undernourishment.	Implement activities to decrease	3.2 # 2	Claim
3)	Contribute to improved health and well-being brought about by reduced levels of fine particulate matter (PM _{2.5}) emissions within households below the baseline	3.9	3.9.1 Mortality rate attributed to household and ambient air pollution	Implemented activities to decrease	3.2 #1	Claim

¹ ACIAR 2013_Promoting diverse fuelwood production systems in Papua New Guinea

	emission level of 4.79 g/kg fuel (or 93.35 mg/min ²)					
4)	Contribute to increasing vocational and relevant skills of local individuals by providing non-formal education and training on issues related to climate change, with specific skill building in operations and surveying activities related to stove distribution and its monitoring under VCS.	4.3	4.3.1 Participation rate of youth and adults in formal and non-formal education and training in the previous 12 months, by sex	Increase	3.2 #3	Claim
5)	Contribute to increasing the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs, and entrepreneurship	4.4	4.4.1 Proportion of youth and adults with information and communications technology (ICT) skills, by type of skill	Increase	3.2 #3	Claim
6)	The project will reduce women and children's drudgery through time savings in reducing time spent on cutting, collecting, and carrying firewood from trees far away from households.	5.4	5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age and location	Implemented activities to decrease	3.2 #3	Claim
7)	Ensure women's full and effective participation and equal opportunities for leadership at all levels of decision-making in the project.	5.5	5.5.2 Proportion of women in the project including managerial roles	Implemented activities to decrease	3.2 #3	Claim
8)	Contribute to increasing access to ICS technology with Burn cookstove installations in around 100,000 households that have been using traditional three-stone fire, over the project lifetime.	7.1	7.1.2 Proportion of population with primary reliance on clean fuels and technology	Increase	3.2 #1	Claim

² <http://catalog.cleancookstoves.org/stoves/215>

9)	With the inception of the project activity, jobs will be created firstly for the distribution of the cookstoves. Further to that and aside from the monitoring done during the verification cycle, the project activity will also conduct regular monitoring of the project sites for additional monitoring and maintenance of the cookstoves and project activity. In turn creating long term jobs for the local community, without prejudice of sex, age, occupation and persons with disabilities.	8.3 8.5	8.3.1 Proportion of informal employment in total employment, by sector and sex. 8.5.1 Average hourly earnings of employees, by sex, age, occupation, and persons with disabilities	Implement activities to increase	3.2 #4	Claim
10)	The project provides ICSs with high thermal efficiency to reduce the consumption of firewood which helps achieve efficient use of an important type of nature resources, non-renewable biomass.	12.2	12.2.2 Domestic material consumption, domestic material consumption per capita, and domestic material consumption per GDP	Implement activities to decrease	2.1 #1	Claim
11)	The project will significantly reduce chemicals (i.e. fine particulate matter (PM _{2.5}) emissions of CO ₂) to the air in order to minimize their adverse impacts on human health and the environment.	12.4	12.4.1 Hazardous waste generated per capita, and proportion of hazardous waste treated, by type of treatment	Implement activities to decrease	4.2 #1	Claim
12)	Contribute to GHG emission reduction through an estimated reduction of 7.2 tCO ₂ e per stove due to the replacement of baseline stoves with Burn Cookstoves over the crediting period.	13.0	Tonnes of greenhouse gas emissions avoided or removed	Decrease	VCS validation report	SD VISta-labeled VCU
13)	Contribute an estimated reduction in removal of woody biomass to the tune of ~3.53 tons per stove per year, from forests surrounding the communities thereby leading to an increase in above-ground biomass in these forests.	15.2	15.2.1 Progress towards sustainable forest management through the increase of above-ground biomass in forests	Implemented activities to decrease	4.2 #1	Claim

2 PROJECT DESIGN

2.1 Project Objectives, Context and Long-term Viability

2.1.1 Summary of Project Sustainable Development Objective(s)

Before the implementation of the project in Papua New Guinea, most of the local people in the island used non-renewable woody biomass for their fuel needs, for cooking with open fire or three-stone fire. The project activity will distribute fuel-efficient ICS to replace the baseline cookstoves in households. Of which the ICS disseminated through this project will replace open-fired wood-fuelled cooking methods.

The project will enable and enhance households to achieve several sustainable development objectives:

The project provides ICSs with high thermal efficiency to reduce the consumption of firewood which helps achieve efficient use of an important type of nature resources, non-renewable biomass. (SDI12.2.2) This in turn impacts the estimated reduction in removal of woody biomass to the tune of ~3.53 tons per stove per year, from forests surrounding the communities thereby leading to an increase in above-ground biomass in these forests. (SDI15.2.1)

In relation, the project activity will reduce women and children's drudgery through time savings in reducing time spent on cutting, collecting, and carrying firewood from trees far away from households. (SDI5.4.1)

The project activity will replace 100,000 traditional cookstoves with fuel efficient improved cookstoves (ICS). The efficiency of changing from a traditional cookstove to a more efficient one will result in the consumption of lesser firewood that is usually purchased at high prices. Hence reducing the economic burden of families under the project activity. (SDI 1.1.1)

The distribution of ICSs will also introduce and increase access the local population to ICS technology with Burn cookstove installations in around 100,000 households that have been using traditional three-stone fire, over the project lifetime. (SDI7.1.2)

It will also contribute to the improved health and well-being brought about by reduced levels of fine particulate matter (PM_{2.5}) emissions within households below the baseline emission level of 4.79 g/kg fuel (or 93.35 mg/min³) (SDI 3.9.1)

³ <http://catalog.cleancookstoves.org/stoves/215>

Further to the above, the project activity will improve food security and nutrition status, particularly for children and women by reducing inadequate cooking, the burden of firewood collection, the time to prepare food. (SDI2.1.1)

With the inception of the project activity, jobs will be created firstly for the distribution of the cookstoves. Further to that and aside from the monitoring done during the verification cycle, the project activity will also conduct regular monitoring of the project sites for additional monitoring and maintenance of the cookstoves and project activity. In turn creating long term jobs for the local community, without prejudice of sex, age, occupation, and persons with disabilities. (SDI8.3.1 and SDI8.5.1)

With the creation of jobs, the project proponent will also carry out training for vocational and relevant skills of local individuals by providing non-formal education and training on issues related to climate change, with specific skill building in operations and surveying activities related to stove distribution and its monitoring under VCS. (SDI4.3.1)

All in all, contributing to GHG emission reduction through an estimated reduction of ~7.2 tCO₂e per stove due to the replacement of baseline stoves with Burn Cookstoves over the crediting period. (SDG 13)

2.1.2 Description of the Project Activity

The Fuel-Efficient Cooking in Papua New Guinea project involves distribution of fuel efficient improved cookstoves (ICS) in Papua New Guinea (PNG). The ICS disseminated through this project will replace open-fired wood-fuelled cooking methods.

The project will be undertaken by Tasman Environmental Markets Asia Pacific (“**TEM**”), during which an estimate of 100,000 cookstoves will be distributed free-of-charge to households in project area. TEM is also dedicated in creating additional value through this project, enhancing the local community’s understanding and awareness health, well-being, climate change and sustainable development.

The primary activity under this grouped project is the installation of ICSs.

Improved Cookstoves (BURN’s Kuniokoa Generation 2 Cookstoves):

The primary project activity is the distribution and installation of BURN’s Kuniokoa Generation 2 Cookstoves, utilizing an air-cooled design in order to keep the inside of the combustion chamber hot and the outside cool, increasing durability, performance and efficiency. Increased efficiency in burning woody fuel would mean less wood being used during the combustion process, encouraging local household to transition away from traditional open fire cooking to cleaner, more efficient cooking solutions.

Training:

In terms of training, TEM will conduct a 2-day course⁴ for its trainers on climate change including the impacts of household pollution on health and well-being, as an introductory background to the efficient cookstove intervention. Training will generate positive impacts on community groups by enabling local community members to build alternative skills and build capacity. These interventions are expected to advance socio-economic status, generate greater community capacity for sustainable livelihoods, and increase climate resilience.

During ICS distribution, these trainers will assist stakeholders in understanding:

1. How the project activity works and its associated socio-economic benefits
2. How to use the ICS
3. End-User Agreement between TEM and End-Users

2.1.3 Implementation Schedule

Date	Milestone(s) in the Project's Development and Implementation
31/01/2023	Consultation with Climate Change and Development Authority (CCDA) on Project Design and Local Stakeholder Consultation
31/01/2023	Consultation with PNG Forestry Authority (PNGFA) on Project Design and Local Stakeholder Consultation
13/03/2023	Local Stakeholder Consultation in Southern Highlands Province
15/03/2023	Local Stakeholder Consultation in Western Highlands Province
17/03/2023	Local Stakeholder Consultation in Jiwaka Province
20/03/2023	Local Stakeholder Consultation in Eastern Highlands Province
	Listing of Project under the VCS Program
	Application of Project under the SDVista Standard
	Project Start date: Project Activities commence with Kuniokoa Cookstove distribution and installation
	Planned time of finishing distribution of ICS
	The 10-year project crediting period

2.1.4 Project Proponent

Organization Name	Tasman Environmental Markets (TEM) Asia Pacific Pte. Ltd.
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⁴ Training course deck available to VVBs on request

Role in the Project	Project Developer
Contact Person	Dr. Gabriel Machovsky Capuska
Title	Head of Science and Sustainability
Address	15 th Floor, 31 Market Street, Sydney, NSW 2000, Australia
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2.1.5 Other Entities Involved in the Project

TEM Asia-Pacific Pte Ltd is the sole entity involved in the project.

2.1.6 Project Type

The project is categorised under type/category as below:

- a) Sectoral scope: 03 - Energy demand
- b) Type: II – Energy efficiency improvement projects

The project is a grouped project activity.

2.1.7 Project Location

The GPA will distribute ICS to communities within the project boundary of PNG (Latitude: 6.3150° S Longitude: 143.9555° E). The first project instance is projected to be within the Highlands Region (Figure 1), and will include the following areas: SHP, WHP, EHP and JP (see Table 1 & Figure 1).

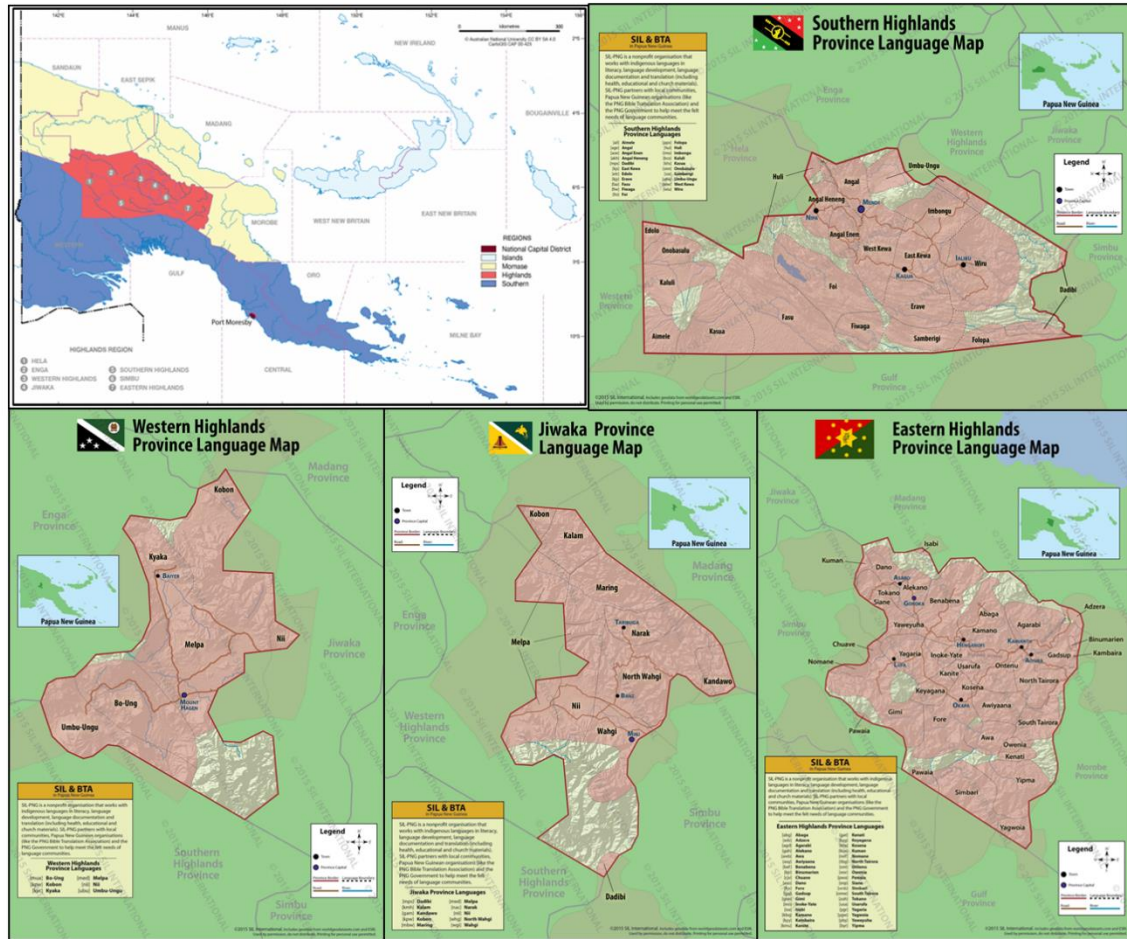


Figure 1. Map of Papua New Guinea including the Highlands Region (top left) and maps of the four different provinces included in the first project instance, Southern Highlands Province (top right), Western Highlands Province (bottom left), Jiwaka Province (bottom center) and Eastern Highlands Province (bottom right). Maps extracted from www.pnglanguages.sil.org

Table 1. The Geographic coordinates of the four provinces involved as the first project instances of the group activity in Papua New Guinea.

S/N	Location	Latitude	Longitude
1.	Southern Highlands Province	6.4179° S	143.5636° E
2.	Eastern Highlands Province	6.5862° S	145.6690° E
3.	Western Highlands Province	5.8310° S	144.1720° E

4.	Jiwaka Province	5.8564° S	144.5218° E
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2.1.8 Baseline Scenario

The conditions prior to project initiation are the continued use of non-renewable wood fuel (firewood) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity.

The pre project devices are inefficient three stone fired or conventional systems lacking improved combustion air supply mechanism and flue gas ventilation system i.e., traditional stoves. The baseline scenario is the same as the conditions existing prior to the project initiation i.e continue the use of traditional cook stoves.

2.1.9 Causal Chain(s)

Please refer to Appendix I for the Causal Chain image.

2.1.10 Threats to the Project

Natural-induced threats

Threat:	Continued deforestation and degradation make firewood an untenable source of biomass.
Solution:	As much as the efficiency of the cookstoves coupled with the imminent scarcity of the wood fuel would support the use of TEM Kuniokoa Cookstoves, there is the potential that as the population continues to expand, there may be a point when firewood becomes an untenable source. In response to this, TEM in country team will train end users to maximise the use of small twigs and branches, crop residues near the local communities they live.

Human-induced threats

Threat:	Stove users' lack of upkeep of stove resulting in either malfunctioning of the stove as designed or lack of stove use.
Solution:	TEM will conduct routine visitations to end-user households at least twice a year in each main geographic focus for mass stove installation to detect defects in stove construction, maintenance, and sub-optimal stove use. This would in turn require TEM PNG Team to hire, train, and supervise users, and to visit each household to inspect their stoves, up to 2 times per year. This dramatically reduces the threat listed above.

Threat:	Failure of Behaviour Change resulting in high levels of non-adoption
Solution:	TEM will continuously research the impacts of behaviour change for successful adoption. Study outcomes, whether formal or informal, are included in future training and community sensitizations to continue to positively, shape knowledge, and enhance positive peer-to-peer influence. Additionally, non-adopters will also be engaged to understand the barriers to success, these outcomes and conclusions are included in future training, and disinformation is countered through community sensitization. To add to the above, TEM will include a utilize Equation 6 of the Methodology, under which the μ_y parameter for the $B_{old,adjusted}$ parameter is listed as 0.1 or 10%. Signifying the fact that 10% of end-users may use both ICS as well as open fire practices.
Threat:	ICS Projects and distributions is not shown to have an increased impact on household uptake.
Solution:	Co-benefits like time savings and increased health benefits aligned with the adoption of ICS work as prominent incentives for increased uptake of these stoves. In response to this, TEM will conduct trainings highlighting how factors such as time savings, health benefits, previous users' reasons for adoption are important factors for large scale adoption.

2.1.11 Benefit Permanence

To ensure the ICS project's long-term viability and the permanence of ICS and other program benefits, TEM will take the following long-term procedures:

- 1) Continuous check on end users to ensure the continued use of ICS- to guarantee this, Implementation and Spot Audit teams will be hired to conduct year-round audits. These teams will be provided with transportation and smart mobile phones that meet the physical mobility and remote connection requirements of the positions. Transportation provided includes bicycles, motorbikes, or motor vehicles, depending on the position.
- 2) Conduct training sessions for health promoters, communities, and individuals, increases the awareness levels of the communities regarding the long-term environmental, health, nutritive, and resiliency benefits aligned with the adoption of ICS. The understanding of these holistic benefits to the stakeholders helps in ensuring long-term sustenance and maintenance of project activities and achieving the intended outcomes.

2.2 Stakeholder Engagement

2.2.1 Stakeholder Identification

Under this project, the stakeholders are identified as local people, communities and or representatives who are directly or indirectly affected by the project, such as end-users, stove manufacturer are identified as stakeholders. The process of identifying stakeholders involves categorizing individuals or groups who are expected to have a direct or indirect impact on the project, or those who have an interest in it. These categories include:

1. Local people, communities and or representatives who are directly or indirectly affected (adversely affected or beneficiaries) by the project or may have an interest in the project.
2. National government officials or National Focal bodies responsible for the project in the host country, for example, Provincial Representatives and Government Officials.
3. Local non-governmental organisations (NGOs), Women Groups working on topics relevant to the project or working with communities who are likely to be affected by the project.

Further to the above, individuals who are interested in the project are encouraged to participate to provide their insights.

2.2.2 Stakeholder Description

The stakeholders identified through the process described in section 2.2.1 above can be generally grouped under three broad categories as follows:

1) **The ICS end-users**

The ICS end-users are directly affected by the project would typically be local people, communities and or representatives who are directly or indirectly affected by the project or may have an interest in the project.

Before the start of the project, users were cooking on traditional cookstoves and not aware about clean cooking technology. Using ICS, less fuel will be consumed to generate equivalent thermal energy as that in the baseline. Reduction in fuel consumption will help beneficiaries in reducing fuel expense and saving time in cooking activities (by faster cooking) as well as fuel procurement.

Additionally, it will reduce the drudgery of the local people through time savings in cutting, collecting, and carrying firewood from trees far away from households as well as cooking over toxic smoky open fires.

2) **Local authorities**

Local policy makers and representatives of local authorities: The project is expected to have indirect benefits for local authorities by improving the livelihoods of the people in the project area and reducing deforestation, thereby promoting overall

development and better living conditions in the region.

The support from local government is very crucial for the implementation of the project. It will provide indispensable information and authorization for the project. It will also help the project implementers to collect feedback from end-users.

3) **NGOs**

Local non-governmental organisations (NGOs), Women Groups working on topics relevant to the project or working with communities who are likely to be affected by the project: These organizations operate within the same geographical areas as TEM's projects. Their involvement would include identifying beneficiaries, managing distribution and logistics, as well as providing training, education, and capacity building.

TEM will collaborate and work closely with local NGOs to make sure that the project will be implemented efficiently.

2.2.3 Stakeholder Consultation

The local stakeholder consultation took place in 4 Highlands Provinces, namely the Southern Highlands Province, Eastern Highlands Province, Western Highlands Province, and Jiwaka Province, on the following dates:

S/N	Province	Date
1	Southern Highlands Province	13/03/2023
2	Western Highlands Province	15/03/2023
3	Jiwaka Province	17/03/2023
4	Eastern Highlands Province	20/03/2023

Prior to the local stakeholder consultations (LSCs), TEM utilized several media outlets to advertise the event to the provinces. These efforts were fulfilled using:

- Newspaper Advertisements
- Letters of Notification to Provincial Governments
- Radio Broadcasts
- Invitation Letters to Churches
- Social Media Posting
- Webinar

All advertising outlets were done in a staggered format with the aim of keeping stakeholders informed and aware of the LSCs that would be happening within their provinces. For exact dates of advertisements being published, please refer to Appendix II: LSC Outreach.

The stakeholders engaged in this project were people or groups of people that were directly or indirectly affected by the project as well as those who were interested in the project and or had the ability to influence its outcome either positively or negatively. They include:

- Representatives from the provincial administration
- Representatives from Women Groups
- Representatives from Churches
- Ward councillors
- Heads of households both male and female
- Key informants such as teachers, community health workers etc.
- Cook Stove (ICS) users (SHP & WHP only)
- Youth leaders and
- Educated elites (lecturers, nursing officers, Health Extension Officers, Police personals etc.)

Southern Highlands Province LSC

A total of 159 people attended this LSC on the 13th of March 2023 held at Kumin Catholic Mission in Mendi, SHP. Including those who commuted into Mendi from various districts, there were public servants such as nursing officers, police, teachers and church elders and pastors present during the consultation.



Figure 2. Demonstration on the use of Improved cookstoves by Ms. Magret Yangpin and Mr. Richard Manda (Field Assistants) during local stakeholder consultation at Southern Highlands Province.

The LSC was facilitated by Mr. Michael Kisombo, the project Manager with Simi Oa Consultancy Services who was engaged by TEM did a power point presentation after welcoming and acknowledging all the stakeholders that were present. The PowerPoint slides were prepared in English, and explained in *tok pidgin*, the common language where most could understand well as those participated came from different local dialects.

Attendees were briefed of the purpose of the meeting, and informed of the:

- Carbon cycle and the important role it played within their forests.
- Findings of the Pilot Project conducted in SHP and WHP
- The Verra and SDVista registration process
- The role of carbon finance in the project activity
- Indicative timeline and implementation plan of the project

Two field assistants were then invited to demonstrate and to explain how the improved cookstove is used using the sample that was exhibited during the LSC. That was further reinforced by one of the users who was also invited to give her experience on how she is currently using the improved cookstove and explained how her and her family are benefiting.

After the presentation, stakeholders were given the opportunity to ask questions and provide their reflections. For the summary of questions asked and subsequent answers provided by TEM, please refer to Appendix III: Stakeholder Feedback.

After the discussions, there were about 29 feedback forms that were received through the feedback forms. These feedback forms may be made available to validators upon request.

Western Highlands Province LSC

A total of 61 people attended this LSC on the 15th of March 2023 held at Kuri Lodge in Mt Hagen, WHP. Participants who attended came from different backgrounds and electorates or districts. The LSC was also covered by Radio Eagle (local radio station) that was present all throughout the presentation.



Figure 3. Demonstration on the use of Improved cookstoves by Mr Tony Yambo (Field Assistant) during local stakeholder consultation at Western Highlands Province.

Prior to the introductions and presentation, Pastor Enoc Wan of Hagen Central – Pentacoastal church opened the session with a word of prayer. Mr. Michael Kisombo then introduced the project and invited his field assistants to introduce themselves. He then proceeded with the presentation using *tok pidgin*.

He gave a brief introduction of the flow of his presentation and explained the purpose of the meeting. He then introduced the project activity before giving an overview of the carbon emissions and the forests' role in restoring carbon and its impact if most forests are destroyed. Following the background of the project, he explained the pilot project conducted in WHP and the results attained from the survey and invited end users (from the pilot project) present to explain how they are currently using the improved cookstoves to date and its impacts on them.

Two volunteers in the group gave their reflections and how they are using the cookstove. Both spoke highly of the impacts the ICS had on their families. Michael then asked Tony (one of the

fields assistants) to demonstrate how the improved cookstove is used. After that presentation, Mr. Kisombo proceeded with the explanation of the Verra registration process and role of carbon finance and continued to explain the large-scale project, its indicative timeline and implementation schedule.

After the presentation, stakeholders were given the opportunity to ask questions and to give their reflections followed by a closing prayer by Pastor Enoc. There were 29 feedback forms received through the feedback forms out of the 61 participants.

Jiwaka Province LSC

More than 90 people attended this LSC on the 17th of March 2023 held at Jiwaka Mission Resort (JMR) in Banz, Jiwaka. During initial consultations to invite the provincial administrator to the LSC, the provincial administrator showed great support for the project and was able to communicate widely to his people to participate. Due to work commitments, he sent his Deputy Administrator to attend the local stakeholder consultation instead.



Figure 4. the Deputy Administrator of Jiwaka Province (Left) and Project Manager, Michael Kisombo (right) and during local stakeholder engagement at Jiwaka Province.

Mr. Michael Kisombo led the discussions with a welcome and thanked everyone who came to Jiwaka Mission Resort for the local stakeholders meeting. He also did the presentation in *tok pidgin* – aided by the slides on power-point that was used in two other provinces. Similar to past LSCs, the topic on carbon cycle and the destruction of the ozone layer was simplified to the level of the participants. This topic was further elaborated on and supported by a few participants who had wider experience working or teaching in schools in this area of specialty.

The pilot in SHP and WHP and the coordinators of these projects, Mr. Patrick Wapiyu and Arnold Terry, were also on hand to explain the results of the pilot study and demonstrate how the improved cookstove was used. After these discussions, Mr. Kisombo continued with the presentations on the purpose of the stakeholder meeting, the verra registration process, carbon

finance and the sustainable goals that this project is meeting. Towards the end, the indicative timeline and implementation schedule was explained before allowing for question-and-answer session.

There were about 29 feedback forms received through the feedback forms out of the 90 registered participants.

Eastern Highlands Province LSC

Around 29 people attended this LSC on the 20th of March 2023 held at Steakhouse Goroka in Goroka, EHP. The low turn-out rate may be attributed to the project being very new to the local stakeholders. Most stakeholders who attended learnt of the event word of mouth or through friends' network and face-book post. Despite a lesser number the discussions were fruitful. Almost all agreed that they see there is no harm and supported the project.



Figure 5. Participants reading the filling up the feedback forms during the Eastern Highlands LSC at Steak House Goroka.

Similar to the first three LSCs, the presentation was conducted by Mr. Michael Kisombo in *tok pidgin* – as the common language where most could understand well. Michael followed the same presentation he did in the other three provinces. The sample ICS that was used for this LSC was displayed and a demonstration was done on how the ICS would be efficiently used. Although it was the first presentation made to the stakeholders in the Eastern Highlands Province, everyone who attended was positive with the project and supported the project activity.

After the discussions, there were about 24 feedback forms received from the 29 participants.

At the end of the LSCs, stakeholders were informed of the project's email (pngcookstoves@tem.com.au), as well as direct contact details to the local project manager and lead project manager in TEM. In all marketing documents, the project email and website were also published, to ensure stakeholders had multiple avenues to reach out to TEM with feedback, allowing for on-going communication.

For more information, a Local Stakeholder Consultation Report is available upon request of the validator.

2.2.4 Continued Consultation and Adaptive Management

Consultation and Adaptive Management will continue throughout the project lifetime as TEM is committed to best meet the needs of the beneficiaries and amplify, where possible, the co-benefits of the project activities as they relate to the SDGs.

Alongside the above, TEM aims to explore additional projects within the region for the implementation of these activities. This is to multiply sustainable development and improve health and economic wellbeing within the project area.

To better adapt to the changing landscape of the project region, TEM will collect inputs from TEM PNG Team twice a year. Interactions may range from informal consultation discussions to more formal surveys with specific questions on use and adoption. Some of the visits are conducted that inquire about likes, dislikes, adoption rates, benefits that TEM may not have identified, and unidentified challenges for the cook.

2.2.5 Anti-Discrimination

TEM will distribute ICSs to local people who uses traditional low efficiency cooking solutions without distinction in genders, races, religions, educational backgrounds, or any other aspects. The PNG Team and TEM are also dedicated to the elimination of discrimination, and as jobs are created during the implementation of this project, TEM will ensure that women, persons with disabilities, and minority will have equal chance to get these jobs.

2.2.6 Worker Training

The implementation of project needs plenty of local people to participate in production, distribution or use steps. They will get relevant skills and sustainable development and global citizenship education through training by project proponent. Besides technical skills, the workers will be trained about the management structure, regulations, and worker rights.

Training about monitoring plan will be provided to TEM's PNG Team, including survey method, data record and analysis. The monitoring plan will be carried out by qualified personnel trained for quality assurance and quality control.

These trainings will be done by TEM over a two-day training course.

2.2.7 Equal Work Opportunities

TEM is dedicated to make sure that all stakeholders, no matter their genders, races, religions, educational backgrounds, or any other aspects, will be given an equal opportunity to fill all work positions. Recruitment advertising for the local villages will be given before the hiring of the full-time and part-time staff for the project. The recruitment advertisement will be provided in Pidgin English for them to understand.

2.2.8 Workers' Rights

TEM and its PNG Team have a labour contract with each worker. Before the workers sign it, the contracts will be explained explicitly to make sure that they could understand their rights and obligations. The contract terms will conform with, and uphold the principles and rights of, work addressed in the Core Labour Conventions of the International Labour Organization (ILO), as well as comply with national and local labour laws within Papua New Guinea.

2.2.9 Occupational Safety Assessment

To mitigate potential occupational safety hazards, TEM makes the effort to contract workers who have lived in the community for several years, thus assisting in mitigating the occupational hazards through community familiarity, language fluency, and being native to the culture. This understanding of traditional values, respect, and working environment in the communities TEM serves to support reduction of safety hazards.

Risks are assessed and mitigated where possible and workers, stakeholders, and others directly involved with the projects are made aware of the potential risks and opportunities to mitigate where possible, yet it should be noted that employment or use of the stove does not increase the safety risks that are already present in the landscape of the country.

2.2.10 Feedback and Grievance Redress Procedure

During the Local Stakeholder Consultations, it was discussed how stakeholders may keep in contact with TEM's country manager, Michael Kisombo. During which, the following solutions were offered:

1. Reaching out to the TEM PNG Team during the twice a year visits to maintain the ICSs.
2. Reaching out to TEM through the project's dedicated email.
3. Reaching out to TEM through the project's website.

The following website of the project activity was available to the stakeholders:
<https://www.tasmanenvironmental.com.au/pngcookstoves/>

The following email addresses will be also used for inquiry or comment on the project:
pngcookstoves@tem.com.au

TEM Country Manager: Michael Kisombo, +675 7262 3864, mkisombo@hotmail.com

The Feedback and Grievance Redress Policy and Procedure are designed and implemented by Tasman Environmental Markets for the proposed high-efficiency wood-burning cookstove project in Papua New Guinea (Figure 6). The primary aim is to lay down the procedures which must be followed to address community and individual grievances related to project stove installation and its operation. These measures are taken to enhance TEM's accountability and transparency and to support the project initiatives.

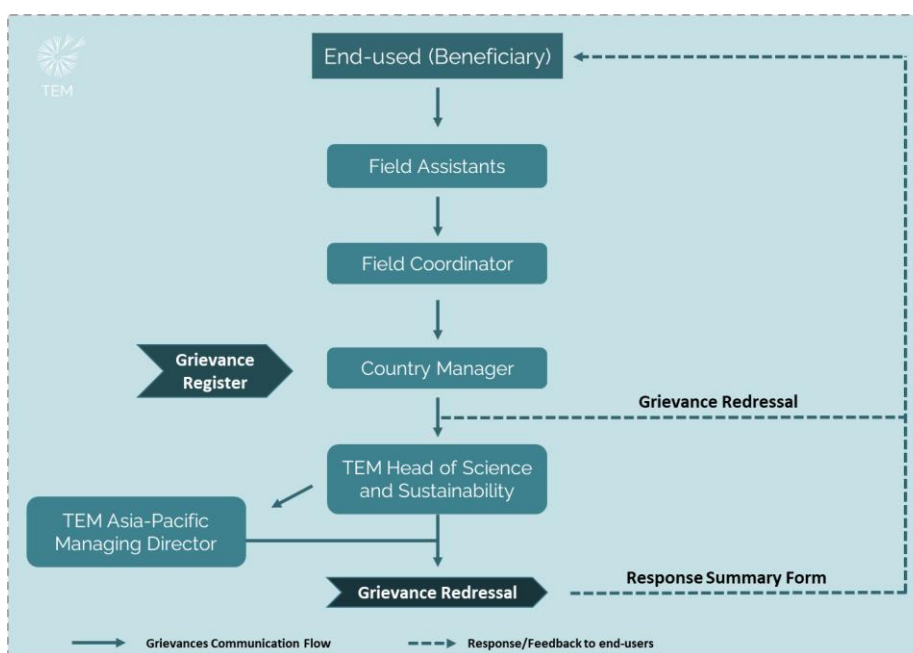


Figure 6. Organizational chart for feedback and grievance redress policy and procedure

2.2.11 Feedback and Grievance Redress Procedure Accessibility

The details on procedure for feedback and grievance have been made public on website of TEM as the following, which is accessible to all stakeholders of the project:

<https://www.tasmanenvironmental.com.au/pngcookstoves/>

Additionally, it will be socialized and publicized to communities throughout the project location in Pidgin English.

2.2.12 Stakeholder Access to Project Documentation

TEM will inform the beneficiaries and their families that they are participating in a project that distributes the Kuniokoa cookstove upon the household's receipt of the ICS. TEM will also inform them of the need to track their successes and monitor key data, such as money spent on wood fuel and/or time spent collecting wood fuel, among other indicators. ICS end-users will also be made aware that to ensure the integrity of the project and programs, auditors, and stove champions, visit the families up to 2 times after implementation.

As for project documents, the full project documentation will be uploaded on VERRA website, and the same ones will be publicized on the TEM website during the whole crediting period of the project as the following linkage:

<https://www.tasmanenvironmental.com.au/pngcookstoves/>

2.2.13 Information to Stakeholders on Assessment Process

TEM and its PNG Team inform end-users that they are participating in a project that distributes the ICS free of charge to improve their respiratory health, the family economy, and the environment. They are informed in advance that the use of ICS generates carbon credits which in turn are used to cover the cost of ICS production and distribution through donation and carbon transfer agreement. To make sure that all the stakeholders know the process of SD VISta project assessment, including the site visits of assessors, TEM and its PNG Team will inform them through the website linkage, phone, email or in person.

2.3 Project Management

2.3.1 Avoidance of Corruption

TEM follows a strict and robust has rules and regulations about the avoidance of corruption, code of conduct, and business ethics of which all staff abide by. These rules and regulations will also be provided to TEM's PNG Team, who should promise to comply with them for cooperation and cooperation with TEM.

Any person or organization which violates the anti-corruption rules and regulations of TEM cannot continue to work for or cooperate with TEM. TEM will ensure that the implementation of project in accordance with all legal requirements and is held to the highest standard of operation.

2.3.2 Statutory and Customary Rights

The project activity involves distribution of ICSs to individual households only and it will not involve any land use or acquisition.

2.3.3 Recognition of Property Rights

All ICSs covered in this project activity are distributed to individual households free of charge. The property right of ICS belongs to end-users while that of carbon credits generated from the

project belongs to TEM. The end-users will sign carbon transfer agreements in the form of End-User Agreements with TEM when they receive ICSs to confirm the property rights of ICSs and carbon credits.

2.3.4 Free, Prior and Informed Consent

The project is voluntarily implemented by TEM, and end-users are free to choose whether they take part in the project or not. Free, prior, and informed consent takes place before distribution through signing of the donation and carbon transfer agreements when the end-users receive the ICSs, which clarify the property rights of the ICSs and the carbon credits generated from the project.

2.3.5 Restitution and/or Compensation for Affected Resources

The project activity involves distribution of ICSs to individual households only and it will not affect any resources.

2.3.6 Property Rights Removal/Relocation of Property Rights Holders

The project activity involves distribution of ICSs to individual households only and it will not lead to any removal of property rights or relocation of property rights holders.

2.3.7 Identification of Illegal Activities

Theft and corruption may be identified during the implementation of the project. TEM and its local partner will avoid any corruption as per Section 2.3.1 of this document. End-users will be reminded to keep their ICSs in safe places to reduce theft when they receive the ICSs.

2.3.8 Ongoing Conflicts or Disputes

The project activity involves distribution of ICSs to individual households only. There is no ongoing or unresolved conflicts or disputes over rights to lands, territories and resources and any disputes that were resolved during the last twenty years.

2.3.9 National and Local Laws and Regulations

There are no laws, statutes, and regulatory frameworks about the application of improved cook stoves in Papua New Guinea households. TEM and its PNG Team obey all relevant local laws and regulations.

2.3.10 Project Ownership

TEM will purchase ICSs and distribute them to end-users free of charge. Before the distribution, end-users and TEM will sign an agreement to confirm that the property right of ICS belongs to end-users while that of the carbon credits generated from the project belong to TEM.

2.3.11 Grouped Projects

The Project is a grouped project activity.

New project activities are assessed and evaluated to determine if they provide significant individual impacts (funding of education), household (cookstove), or community level (training on climate change).

For the inclusion of new project activity instances i.e., ICS, the project proponent shall ensure that it meets the eligibility criteria below – per Appendix 1 – Grouped Projects (GP) of the SD VISta Standard.

S/N	Criterion	Compliance
1	Adopt and implement the project activities in the same manner as specified in the project description.	New project activity instances will be implemented in the same manner as described in the project description, extending benefits, and reinforcing project stove adoption.
2	Where appropriate, meet the applicability conditions of the SD VISta asset methodology applied to a project.	The project activity, as well as new project instances will apply the SD Vista asset methodology and adhere to the criteria set out in the methodology.
3	Are subject to the same scenarios at project start with respect to stakeholders' well-being as determined for initial project instance(s), where (per Section 2.1.5.2 above) the project must meet the criteria of Section 3.1 above.	New project activity instances will have the same baseline scenario, which was available during the start of the project, i.e., usage of BURN's cookstoves in the households. Also, the instances will be designed to maximize the intended impact and preserve well-being, with monitoring and ongoing stakeholder consultations to ensure impact.
4	Are subject to the same scenarios at project start with respect to natural capital and ecosystem services as determined for initial project instance(s) where (per Section 2.1.5.2 above) projects must meet the criteria of Section 3.2 above.	New project instances will have the same scenario as determined in the baseline scenario. All the new instances will also have households relying on wood fuels for meeting their daily cooking energy demands.

5	Are subject to the same processes for stakeholder engagement described in the project description.	The new project activity instances will be detailed during continuous stakeholder consultations, refresher trainings for Implementing Partners and feedback consider in secondary project activity design to maximize the intended impact and preserve wellbeing.
6	Are subject to the same processes for respect for rights to lands, territories and resources including free, prior and informed consent – described in Section 2.4 above.	The outlined Project Management (Section 2.3) will apply to each additional project activity.
7	Have similar monitoring elements to those set out in the project description.	Additional projects will have equivalent monitoring plans and procedures as described in Section 3.3.

3 BENEFITS FOR PEOPLE AND PROSPERITY

3.1 Condition of Stakeholders at Project Start

(a) The ICS end-users

Papua New Guinea's total population of 8.8 million people consumes well over 13 million m³ or 9.4 million tonnes of firewood per year and in doing so contributes significantly to Papua New Guinea's Greenhouse Gas (GHG) emissions⁵. As per the IRENA report of 2013, over 90% of the population occasionally or regularly uses firewood for domestic and commercial cooking on open fires in Papua New Guinea. However, firewood uses vary from area to area or region by region. For example, firewood consumption in the coastal regions may not be same up in the highlands where the altitude is high. Firewood use in the highlands is extremely higher than any other regions. As discussed by Nuberg, 2015, "fuelwood (firewood) is the primary energy source for cooking and heating especially in the highlands where over 40% of the population lives". Therefore, firewood collection and consumption in the highlands is higher than in the lower land areas as the highlanders use firewood not only for cooking but also for heating, lighting, feasting, storytelling and so on.

(b) Local authorities

Local officers welcome the project because the implementation of project will improve the living standard and economic situation within project boundary.

(c) NGOs

There are various NGOs dedicated to help local people in different ways in Papua New Guinea and are experienced in operating different types of collaborative projects. These NGOs cooperate with TEM to make sure that the project goes well.

3.2 Expected Impacts on Stakeholders

Impact #1	Access to ICS
Type of Impact	Positive, actual, direct

⁵ <https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2013/Sep/Papua-New-Guinea.pdf?la=en&hash=3E847FD95A91ADAA4CC34614F7A325F80CE36D39>

Affected Stakeholder Group(s)	ICS end-users
Resulting Change in Well-being	Reduce inadequate cooking, the burden of firewood collection, the time to prepare food, the cost to buy firewood, improved health and well-being.

Impact #2	Improved Health Status
Type of Impact	Positive, predicted, direct
Affected Stakeholder Group(s)	ICS end-users
Resulting Change in Well-being	Reduce the local people's exposure to high PM2.5 and high CO due to higher efficiency of combustion.

Impact #3	Less time spent on unpaid domestic and care work
Type of Impact	Positive, predicted, direct
Affected Stakeholder Group(s)	ICS end-users
Resulting Change in Well-being	Reduce the time spent on firewood collection and cooking for people, especially for women and children.

Impact #4	Increased Income
Type of Impact	Positive, actual, direct
Affected Stakeholder Group(s)	Employees hired for Project Activity

Resulting Change in Well-being	During the project crediting period, TEM will oversee maintenance of cookstoves and the monitoring plan, which will require the need to hire local people.
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Impact #5	Trainings imparted on climate change, project implementation and monitoring procedures
Type of Impact	Positive, actual, direct
Affected Stakeholder Group(s)	Local community, Beneficiaries, TEM PNG Team and Staff
Resulting Change in Well-being	Training and skill development related to community engagement, survey conduction, technical trainings like conducting Water Boiling Tests (WBT) will be provided to many stakeholder groups which is envisaged to empower their lives by not only providing long-term employability but also through increased awareness levels regarding issues related to climate change, social equity.

Impact #X	Improved Economic Outcomes
Type of Impact	Positive, actual, direct
Affected Stakeholder Group(s)	ICS end-users, ICS end-users' Families
Resulting Change in Well-being	TEM predict money and time savings with fuel efficiency reducing costs of firewood purchases and/or time savings on the collection of wood.

3.3 Stakeholder Monitoring Plan

The project activity distributes ICSs to households, which will have a net positive impact on overall well-being of the people in the project area. To account for these positive impacts, a monitoring plan is needed to identify the monitored stakeholder groups, the types of measurements, the sampling methods and the frequency of monitoring and reporting.

TEM and its PNG Team will collect data through interviews, surveys, direct observations, and group discussions about stakeholders' financial, health and employment records. TEM's PNG Team are experienced in facilitating the operations of these projects, and TEM has also trained them to make sure they follow closely to the monitoring plan. They will collect primary information through regular visits and interviews with the end-users and other stakeholders. Direct observations and group discussions will be held at times. The data from these interactions will be compiled into reports and submit to the TEM. To make sure the data is credible, the data may be cross checked by a third-party or TEM.

Data collected will be stored on a storage cloud.

Initial Visit to Project Area

Prior to the implementation of the project, TEM PNG Team will conduct a one-time identification and sensitization visit to local villages. During which, the PNG Team will identify and visit provinces which are suitable for the project activity. This will include having discussions with the leaders of provinces and stakeholders.

They also communicate with other stakeholder such as local officers and NGOs, and search for their support and cooperation. In discussion with stakeholders, they explain the project and its benefits on health, nutrition, and climate change. Any questions raised are recorded and responded. This will be done in the form of a local stakeholder consultation, that will later feed into the project design documents.

After the project plan is accepted, a community-wide training session is held to introduce the project to all interested households. The training includes information on the multiple benefits and how they can participant in the project.

Distribution Visit

When distributing ICSs to end-users, trained TEM staff explain how to use and maintain the ICSs. TEM's PNG Team will respond to any doubts or questions and leave a telephone number so end users can contact TEM or the PNG Team when needed. Before completing the visit, the PNG Team ensure that the end-users are capable to prepare meals on their new ICSs.

Checking Visits

TEM's PNG Team will conduct follow-up visits with households after the distribution. This is the most critical visit because during this time that frustration with the ICS may arise. Villages where the project has been implemented will be chosen randomly. The staff will help end-users facilitate the adaptation, provide support to the women who cook, and solve emerging problems. Feedbacks from end-users will be collected. TEM and its PNG Team will analyse these feedbacks and improve the project activity accordingly when applicable, increasing the adaptation of the community to the project.

Verification visit

During each Verification period, TEM and/or the PNG Team will conduct a random sampling survey on the implementation of the project, including all the effects identified in the project's causal chain related to stakeholder well-being. End-users' using experience and stove condition will be monitored and recorded in the questionnaires. Other stakeholders' opinion and suggestion will also be recorded.

3.4 Net Positive Stakeholder Well-being Impacts

The inception and distribution of ICSs undertaken by the project activity will provide a basic service necessary to lead to a healthy and productive life, including saving time and money for wood fuel at the household level. The project will reduce the drudgery undertaken by them especially for women and children through time savings in cutting, collecting, and carrying firewood from trees. The project will also improve food security and nutrition status by reducing inadequate cooking.

Through the development of the project activity, local stakeholders will yield the following benefits:

1. Reduced Exposure to high CO, PM 2.5 levels and other harmful fumes discharged through inefficient combustion of woody fuel.
2. A higher proportion of population will have access to clean cooking technology for meeting their daily cooking needs.
3. Families will have a greater capacity to save money used for buying wood fuel due to the reduced demand of fuel wood for cooking purposes.
4. Less time will be spent on hard menial, unpaid and dull work of collecting wood fuel and household chores like cooking.
5. Local individuals will be trained and have better vocational skills enhancing their long-term employability and influencing their overall social and economic wellbeing.

4 BENEFITS FOR THE PLANET

4.1 Condition of Natural Capital and Ecosystem Services at Project Start

Papua New Guinea's total population of 8.8 million people consumes well over 13 million m³ or 9.4 million tonnes of firewood per year and in doing so contributes significantly to Papua New Guinea's Greenhouse Gas (GHG) emissions⁶. As per the IRENA report of 2013⁷, over 90% of the population occasionally or regularly uses firewood for domestic and commercial cooking on open fires in Papua New Guinea. However, firewood uses vary from area to area or region by region. For example, firewood consumption in the coastal regions may not be same up in the highlands where the altitude is high and temperature is low (Hanson, et al., 2001)⁸. Firewood use in the highlands is extremely higher than any other regions. As discussed by Nuberg, 2015⁹, "fuelwood (firewood) is the primary energy source for cooking and heating especially in the highlands where over 40% of the population lives". Therefore, firewood collection and consumed in the highlands is more than the low land areas as the highlanders use firewood not only for cooking but also for heating, lighting, feasting, storytelling and so on.

However, there is little documented understanding of firewood usages, its implications and consumption rate available in the country except a few studies conducted by the Australian Centre for International Agricultural Research (ACIAR) 2013, Murphy (2009) and Page et al. (2016). There is also limited literature to understand how much firewood is used for cooking, heating, and other purposes per household especially across the Highlands region of Papua New Guinea. According to the available literature, Murphy (2009) estimated the average firewood consumption for the Eastern Highlands Province as 3.64 tonnes/household/year, ACIAR (2013) proposed that 7.51 tonnes/household/year was the average firewood consumption nationwide based on data collected from the Highlands region (Eastern Highlands Province, Western Highlands Province and Chimbu Province), National Capital District and Morobe Province, and Page et al. (2016) estimated an average of 11.10 tonnes/household/year for the Morobe Province.

4.2 Expected Impacts on Natural Capital and Ecosystem Services

⁶ <https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2013/Sep/Papua-New-Guinea.pdf?la=en&hash=3E847FD95A91ADAA4CC34614F7A325F80CE36D39>

⁷ IRENA (International Renewable Energy Agency), 2013. Renewable energy opportunities and challenges in the Pacific Islands Region

⁸ Hanson, L., Allen, B., Bourke, R. (2001). Papua New Guinea Rural Development Handbook. Canberra, Australian National University

⁹ Nuberg, I. K. (2015). Developing the fuelwood economy of Papua New Guinea. *Energy for Sustainable Development*, 24, 9-18.

Impact #1	Reduced demand for large diameter non-renewable firewood from live trees due to only needing finger-sized woody biomass, i.e., twigs and crop residues
Type of Impact	Positive, actual, direct
Affected Natural Capital and/or Ecosystem Service(s)	Live plant biomass, Biodiversity and species Richness, Soil and Water Conservation
Resulting Change in Condition	By slowing the consumption of woody biomass for cooking purposes and by eliminating the need for large diameter woody biomass due to the technology in the ICSs, the resulting drop in demand slows deforestation, thus increasing the above-ground biomass in the adjoining forests near the project area. Reduction in unsustainable harvesting of plant biomass would in turn reduce soil erosion and result in improvement in biodiversity and species richness of the implementation area.

Impact #2	Avoided emission of GHGs made possible by using ICSs
Type of Impact	Positive, predicted, indirect
Affected Natural Capital and/or Ecosystem Service(s)	Air quality in the households as well as surroundings
Resulting Change in Condition	The project stoves require less fuelwood for producing the same amount of thermal energy, therefore, resulting in less carbon dioxide emission in the atmosphere. As carbon dioxide is a greenhouse gas, its reduced emission translates to a positive impact on climate change.

4.3 Natural Capital and Ecosystem Services Monitoring Plan

To monitor the impact on natural capital and ecosystem services, TEM PNG team will track:

1. The amount of non-renewable biomass used to meet thermal energy needs for cooking using ICS.
2. Survey the difficulty in collecting firewood, including distance and time.
3. Official data about forest resource, including local, national and international.

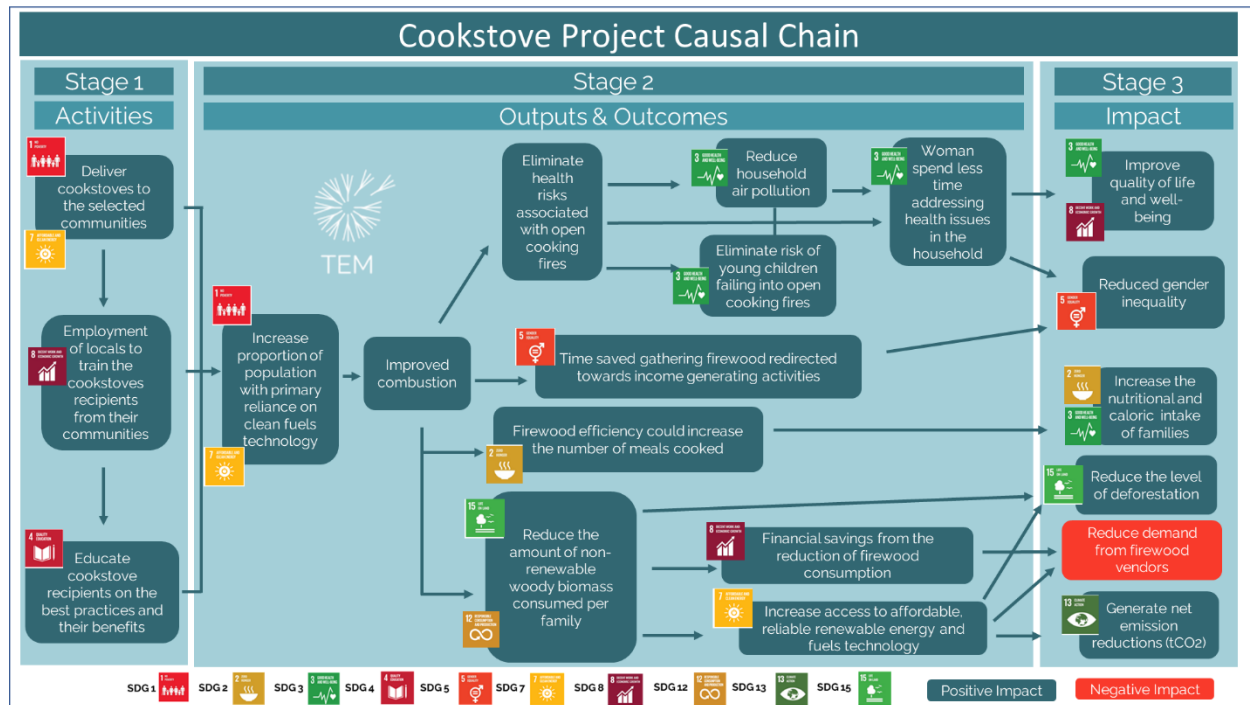
The above data will be collected through survey and fuel consumption test, which is the responsibility of TEM PNG team. TEM will train its PNG Team on the monitoring plan and supervise their work. The data may be cross checked by a third-party. The end-user households where the survey and test are conducted will be chosen through random sampling. The data from test and survey will be recorded, analysed and reported by TEM. The monitoring and reporting will be conducted each time when verification of SD VISta is conducted.

4.4 Net Positive Natural Capital and Ecosystem Services Impacts

By replacing traditional low efficient three-stone fires with high efficiency improved cookstove in households, the project will increase energy efficiency resulting in less firewood combustion, thus generate net GHG reductions.

The average annual GHG emission reduction from the project is expected to be **572,936 tCO₂e**. The crediting period is expected to be 10 years. Hence the total GHG emission reduction is **5,729,361 tCO₂e**. The project will save **~3.53t** non-renewable biomass every year and it will result in a significant reduction of deforestation.

APPENDIX I: CAUSAL CHAIN



APPENDIX II: LSC OUTREACH

1. Newspaper Advertisements:

There were two newspapers published nationally in PNG. As the project activity would be a grouped project activity, TEM saw the importance of announcing the to the public and informing stakeholders of the project, as well as brief them on the upcoming local stakeholder consultations happening in the Highlands Provinces. The following table shows the date of when these advertisements were published in the newspaper:

S/N	Publishing House	Advertisement Published on: (date)	Supporting Documentation
1	Post Courier Newspaper	14/04/2023	Invoices from respective newspapers for publishing, picture of advertisement in newspapers

2. Letters of Notification to Provincial Governments:

Letters of Notification were also hand delivered to provincial government administrative assistants to brief them on the project activity as well as the consequent co-benefits to their local community that came with the project. Together with the Letter of Notification, a project brief was also attached to aid their understanding. The following table shows the dates of when the letters of no-objection were received from the provincial governments:

S/N	Province	Document Received on: (date)	Supporting Documentation
1	Western Highlands Province	13/02/2023	Letter of No-Objection
2	Jiwaka Province	15/02/2023	
3	Eastern Highlands Province	16/02/2023	
4	Southern Highlands Province	08/03/2023	

During TEM's consultations with the provincial governments, the administrative assistants expressed a high level of interest in the project and were supportive of its implementation.

3. Radio Broadcasts

To further TEM's outreach into the Highlands Provinces regarding the LSCs, radio broadcasts were utilized. All radio broadcasts were done by the National Broadcasting Commission which then broadcasted TEM's announcements to individual provinces at a frequency of 1 time a day, over 10 days. Each province-level broadcast would then inform the specific province of the date, location and time of

which the LSC was happening within their area. The following table shows the start and end dates of these broadcasts:

S/N	Province	Start of Broadcast: (date)	End of Broadcast: (date)
1	Western Highlands Province	29/02/2023	12/03/2023
2	Jiwaka Province	29/02/2023	12/03/2023

The radio broadcast could only be done for 2 highlands provinces as it was found that the radio stations for the remaining two provinces had been shut down.

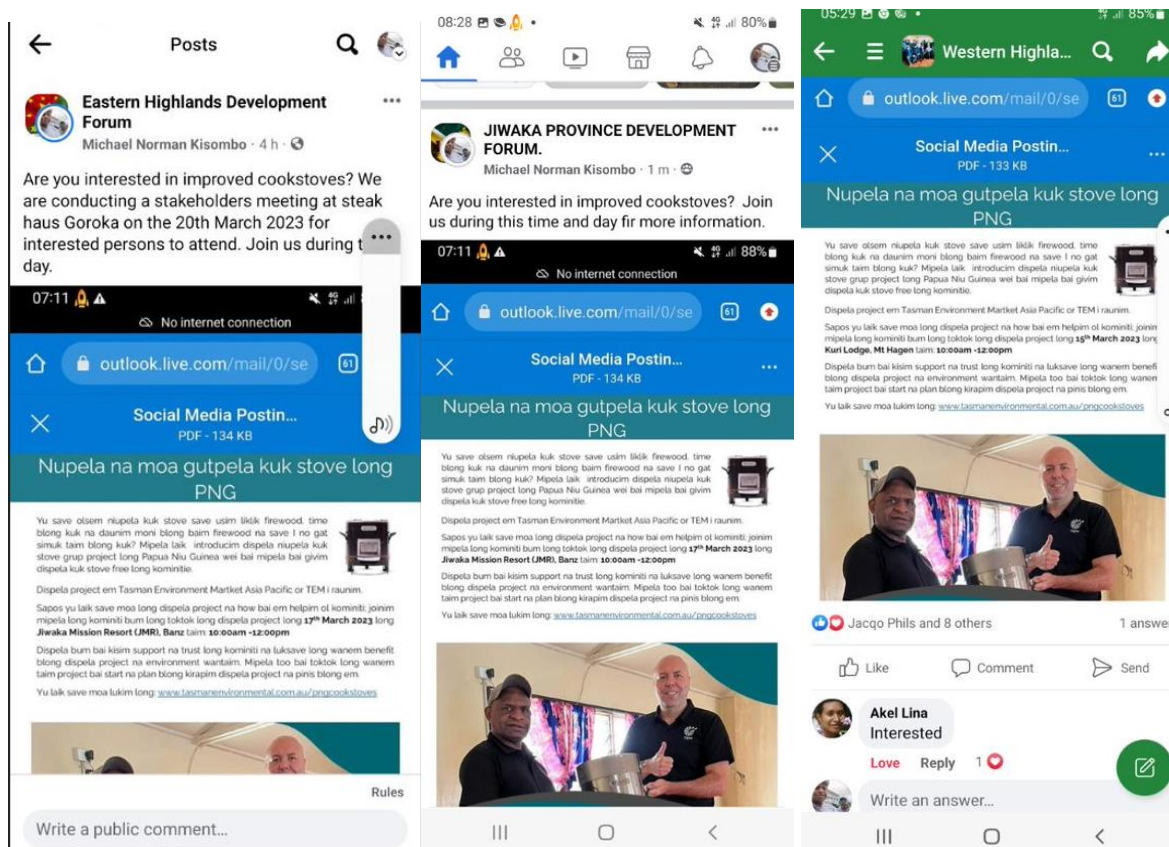
4. Invitation to Churches

TEM is aware that churches and church leaders play an important role in the local community within PNG. As such, invitations to churches were also hand delivered to churches within each province to invite church leaders and elders to participate in the LSC, as well as spread the word about the importance of the LSC within their community by word-of-mouth. The following table shows the dates these invitations were delivered to churches within the provinces:

S/N	Province	Delivery of Invitation: (date)	Supporting Documentation
1	Southern Highlands Province	10/02/2023	Acknowledgement of Receipt by Church Bodies: SHP
2	Western Highlands Province	15/03/2023	Acknowledgement of Receipt by Church Bodies: WHP, Jiwaka, EHP
3	Jiwaka Province	17/03/2023	
4	Eastern Highlands Province	20/03/2023	

5. Social Media Postings

TEM also understood that the majority of the stakeholders received news of events happening within their area through social media. For this, flyers depicting the details of the LSC, and introduction of the project were developed and posted on each province's dedicated Facebook development forums to further boost stakeholder awareness of the LSC. Postings of these flyers started on 13/03/2023.

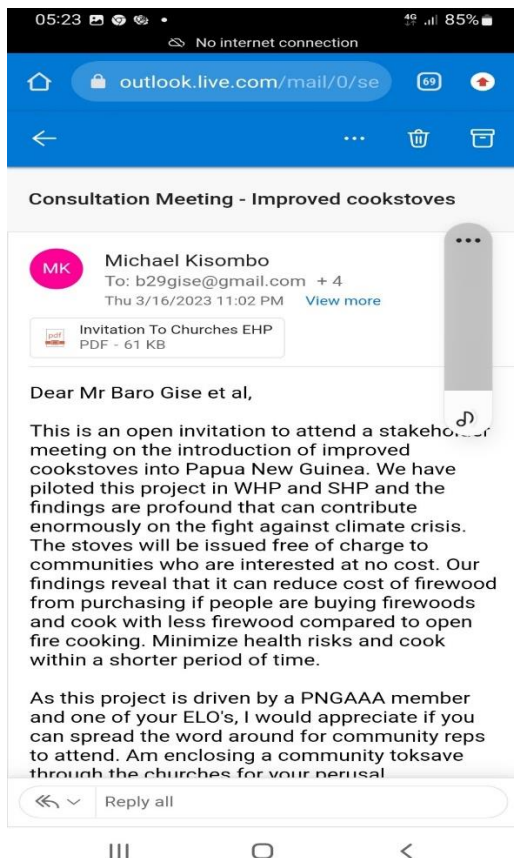
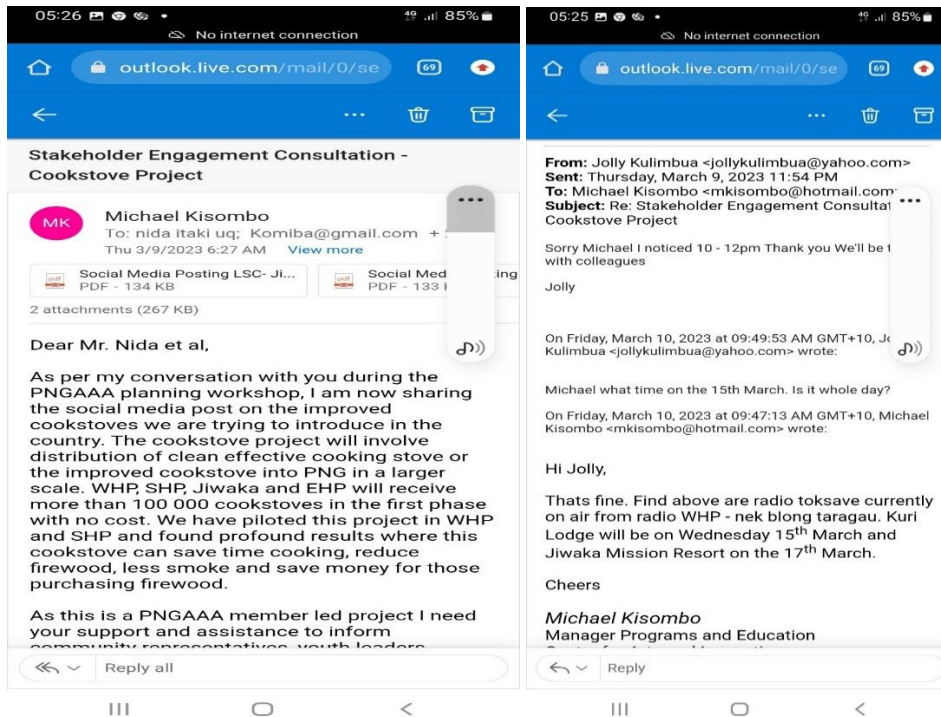


6. Email to PNG Australia Alumni Association (PNGAAA) members

From within TEM's local partner's connections, TEM also reached out to members from WHP and EHP via email. The table below shows the dates these emails were sent out:

S/N	Province	Date of Email Sent: (date)
1	WHP	09/03/2023
2	EHP	16/03/2023

Screenshots of the emails sent out can also be found below:



APPENDIX III: STAKEHOLDER FEEDBACK

1. Southern Highlands Province:

Questions:

S/N	Question	Answer
1	Ms Julie Ponaria (a teacher from SHP) enquired that there are many big countries contributing to the destruction of the ozone layer and now trying to give (us) cookstoves to stop carbon emissions. She asked if there is a story behind this small cookstove?	Yes, there are many industrial countries that are destroying the ozone layer because they have big factories and industries. We (people of PNG) also do so when cutting down trees for farming and cooking etc... This project is to reduce carbon emissions by using less firewood for daily needs. Hence, saving trees that consume carbon and restore them. That is the story behind this project.
2	Ps Joseph Yapasi of Tente Ward 2 commented that 50 000 cookstoves is not sufficient to cater for all households in Southern Highlands and there may be risks when distribution. Therefore, he suggested that distribution be more targeted. E.g distribute through church networks or distribute to a particular district first and later target other districts.	It was noted for TEM to decide.
3	Ms Regina Kamerepa pointed out that since the cookstoves are going to be issued free, people are likely to rush to get one. Hence, can also steal or pull them from the distributors during the distribution. And asked how TEM would manage that scenario?	This is a threat we are aware of and are working on a strategy on how this can be mitigated during distribution. Currently, it is planned that security will be present during the distribution process. End-users will also be advised to keep their cookstoves in a safe place to deter theft of the device.

Reflections:

S/N	Reflections
1	A youth rep commented that the level of presentation was too technical and high given the understanding level of the communities and stakeholders present asked if the presentation can be more tailored to their understanding level.
2	A youth rep commented that the level of presentation was too technical and high given the understanding level of the communities and stakeholders present asked if the presentation can be more tailored to their understanding level.
3	A youth rep commented that the level of presentation was too technical and high given the understanding level of the communities and stakeholders present asked if the presentation can be more tailored to their understanding level.
4	Mr. Peter Wari who is in charge of climate change in the province acknowledged the project as it was really going to benefit the community. He further stressed about the importance of the project activity and encouraged the stakeholders to support the project.
5	Most stakeholders mentioned that there is an increase in the usage of firewood and most of their trees (caussarina/year) are diminishing because people are either chopping them to sell as there is a high demand or for household use. Therefore, the introduction of such cookstoves will be relieving for them.
6	Most stakeholders mentioned that there is an increase in the usage of firewood and most of their trees (caussarina/year) are diminishing because people are either chopping them to sell as there is a high demand or for household use. Therefore, the introduction of such cookstoves will be relieving for them.
7	Another youth mentioned that he provides firewood for three separate houses as most are widows who have no husbands. With the new cookstove, he believes will relieve him from time and struggles he is facing foraging for firewood for all he's mothers.
8	A Health worker emphasized that inhaling of smoke using normal open fire cooking is contributing to a lot of respiratory illnesses and with the introduction of the improved cookstove she believes will minimize such especially mothers and girls.
9	Besides the above reflections and questions, there were quite a good number of people giving positive comments and supported the project activity. They even asked to increase the volume from 50 000 cookstoves to enough to cater for all households the province.

2. Western Highlands Province:

Questions:

S/N	Question	Answer
1	Mr. Frank Kuri Cr from Mul Baiyer asked, since it's going to affect the rural communities, can the councilors get involved?	Yes, Councilors will get involved to identify heads of households but we do not want this project to be

		politicized, either be councilors or local MPs.
2	Mr. Nida L Taki, a lecturer at the Agriculture College, Mt Hagen asked if there are further studies made to determine the level of smoke that is given out during normal open fire cooking to see how much carbon emissions are produced.	That's an interesting question. However, this project used observation method to observe the level of smoke when cooking both during the control cooking test and open fire cooking.
3	Mr Dickson Wia stressed that firewood is not only used for cooking in the highlands but for other purposes too. Will this cookstove provide for all?	Yes, we use firewood for many other purposes like for heat, light, storytelling etc. This cookstove is for cooking and it can also heat up the house.
4	Rex Parak enquired if the design of the cookstove can be made locally?	The brand and model of cookstove was chosen specifically for its high efficiency. TEM will consider a local brand in the future if it fits the specifications required to yield the highest efficiency in the combustion of firewood.

Reflections:

S/N	Reflections
1	The District Administrator for Dei District thanked the team for introducing this new innovative idea of cooking as people in his electorate are running out of firewood. He welcomed the project initiative and asked to include all rural communities in his electorate.
2	Those who came from the urban centers also requested that they should be given priority as they are using a lot of money to purchase firewood and its already becoming a burden.
3	One of the participants mentioned that the project is very beneficial as it saves time foraging for firewood and a financial relief in buying firewood.
4	A youth mentioned that this project will benefit the youth as he sees that firewood are already running out.

5	Another supported by saying, the initiative taken by TEM is indeed very vital to reduce carbon monoxide as well as relieving economic burden and reduce health related issues. She suggested that it would be good to involve the councilors.
6	One said, this project must not stop after 10 years. It must continue till the world ends.
7	A health worker supported the project by saying, as a health worker this project if rolled out effectively will help reduce the number of sicknesses like pneumonia, chronic obstructive lung disease and eye diseases as well.

3. Jiwaka Province:

Questions:

S/N	Question	Answer
1	Mr. Lumun Mingal asked how can PNG be compensated well since we are less carbon emitters yet are asked to further reduce the emissions through such cookstove project?	This cookstove project aims to contribute to the wellbeing of the stakeholders in PNG by lessening the burden of purchasing or collecting firewood and improving the respiratory health of end-users exposed to hazardous smoke from the burning of firewood. Cookstoves being distributed for free is paid for through carbon finance and is one method of compensating stakeholders using the carbon credit scheme.

Reflections:

S/N	Reflections
1	A local landowner thanked TEM for this initiative and welcomed everyone to JMR. Having said that he suggested that this project be based there as it is the model village for Jiwaka and must be made the main distribution point.

2	Another said, there are many NGOs like IOM, UNICEF, etc. who come and deliver similar community-based projects but we the people seem to sell those items. He was worried the cookstoves might be sold in a similar manner and appealed to other participants to discourage such practices.
3	A female teacher mentioned that as a teacher she moves from one area to another with kerosine stoves and with the introduction of the improved cooking stove, hers and other teachers would benefit greatly. Not only that but emphasized that they usually ask their students to bring firewood for them once a week and this project will ease that burden.
4	Another person suggested that every household in Jiwaka must receive a cookstove as they are facing difficulties in looking for firewood.
5	Sammy Buka of Banz mentioned that the project is a bottom up where local communities will benefit. He said it's a sickness free project.
6	There were other supportive comments like, TEM must continue this project to other provinces in PNG, delivered through churches, schools and LLG wards.
7	When asked if there are risks involved like those of you selling firewood, would this project affect you? Everyone said no, not at all as they see there is a lot of benefit to the rural communities.
8.	Someone also wanted more than 50 000 cookstoves in Jiwaka as they believe they are good people.

4. Eastern Highlands Province:

Questions:

S/N	Question	Answer
1	Mr Rex Yagusa mentioned that there is a high possibility of the cookstoves being stolen by thieves or destroyed by natural and man-made disasters and asked in such a case, how can we register these complains or cases?	Currently, it is planned that security will be present during the distribution process. End-users will also be advised to keep their cookstoves in a safe place to deter theft of the device.

Reflections:

S/N	Reflections
1	One of the ladies (video graphed) mentioned that this cookstove will lessen the burden for those mothers in the rural setting. She said if they don't sell, there is no money to purchase firewood so they would use plastics and cardboard to cook. This project will really relieve those struggling mothers living in the urban areas.
2	Another suggested having the distribution point centralized and allowing only genuine households to collect. Later field officers can be sent to their locations for registration such as GPS points etc.
3	Someone in the group mentioned that economic benefits are huge as they will be working as the project team to distribute.
4	Those who attended expressed support for the project and demanded if more is distributed in EHP given their population density.
5	No risks or harm mentioned when implementing this project.
6	One participant mentioned that tree-cutting is happening almost every day. More supplies to meet population will help reduce carbon emission.